

AVI MEHTA

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WORK EXPERIENCE

Charlotte, NC
Dec 2024 - Present

Truist

Data Scientist | Treasury Data Office

- Led development of data infrastructure and feature pipelines processing 2TB+ of monthly data for a \$550B+ banking portfolio, enabling predictive risk models used in regulatory stress testing and enterprise-scale financial workloads
- Engineered General Ledger-to-portfolio reconciliation algorithms into a reusable framework, cutting manual validation effort by 30% and improving high-integrity datasets for model training, evaluation, and downstream analytics
- Partnered directly with risk and compliance stakeholders to convert ambiguous requirements into interactive dashboards, turning complex model output into decisions leadership could act on and improving audit transparency
- Ran statistical trend analysis and anomaly detection across 500K+ loan accounts monthly, surfacing exceptions and edge cases that fed credit risk and early-warning models

June 2023 – Nov 2024

Data Scientist | Truist Audit Services

- Developed anomaly detection models for AML ops against noisy transaction data, improving high-risk entity identification by 9% without increasing false-positive rates
- Architected a random sampling framework and orchestrated scalable data pipelines in Python, cutting manual processing by 85%
- Built predictive models forecasting Adjustable-Rate Mortgage portfolio performance, reducing overall risk exposure by 12%
- Accelerated compliance reviews by 30% through a fuzzy-logic inference system that reconciled OFAC and transaction data

Blacksburg, VA

Virginia Tech Department of Computer Science

Aug 2022 - May 2023

Undergraduate Teaching Assistant | CS3724 Human Computer Interaction

- Held weekly office hours, assisted the professor in writing test and homework material, and graded exams of 150+ students
- Led discussion sections of 48 students weekly, working through problems interactively

Ann Arbor, MI

Blankly Finance (Backed by Contrary Capital and Rough Draft Ventures)

Oct 2021 - May 2022

Software Engineer Intern

- Engineered scalable user registration flows using React, Flask, and Firebase, with full test coverage using Pytest and Cypress
- Increased Blankly supported exchange integrations by 50% leveraging RESTful APIs and web sockets
- Built and deployed trading bots using the Blankly API, leveraging Docker and AWS EC2, to allow users to simulate trades

Blacksburg, VA

The Consulting Group at Virginia Tech

Oct 2020 - May 2022

Technology Consultant

- Constructed a cybersecurity system for smart vehicles and conducted A/B testing to evaluate and mitigate vulnerabilities
- Drove M&A due diligence for a fintech merger, emphasizing WealthTech, blockchain, and digital banking markets
- Orchestrated a 5G adoption and utilization roadmap for a healthcare client leveraging scientific + market research

SKILLS

Languages:	Python, Java, JavaScript/TypeScript, SQL, C/C++, SAS
Technologies & Frameworks:	React, Flask, Node, Next.js, Postgres, MongoDB, GraphQL, Hadoop, FastAPI
Libraries:	TensorFlow, Pandas, PyTorch, NumPy, Matplotlib, Scikit
Infrastructure:	Linux, Git, Docker, Kubernetes, AWS, GCP

PROJECTS (View all on avimehta.info)

Nestor AI

- Launched a no-code agentic AI platform for creating, deploying, and monitoring AI agents with real-time streaming
- Optimized token costs to near-constant via sliding-context windows and async summarization while maintaining output quality
- Designed a reusable agent orchestration layer with multi-provider LLM routing, failover, deterministic fallbacks, and encrypted tenant isolation, supporting 500+ agent runs across varied use cases
- Led 0-1 product development across architecture, UX, deployment, and iteration for scalable rollout

Kernel Mate

- Shipped an LLM-powered terminal assistant as a zero-dependency, cross-platform CLI on PyPI, improving command accuracy from 70% to 95% via context-aware prompt engineering
- Cut response latency from ~2.2s to under 100ms (cached) using SHA-256-based context-aware caching
- Designed a safety-first execution system with risk detection, user confirmation, and automatic undo generation

EDUCATION

May 2023

Virginia Tech

B.S. in Computer Science, Cum Laude